Lorenzo Celli

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Zürich
Italian, B permit

Computer scientist with a strong interest in computer graphics, computational geometry, computer vision, and machine learning. Originally from Bologna, now based in Zürich where I am pursuing my Master's degree.

Professional Experience

Research Assistant *ETH (Safespace Research), Zürich* 2025 – Present

- Implemented a Graph-RAG system for enhanced LLM generation targeted at mental health support.
- Developed the frontend of a mobile application for iOS and Android with React Native, Xcode, Android Studio. Developed a REST API with Django and Django REST Framework. Deployed with Docker on Azure.

CAD Software Engineer *devDept, Bologna* 2020 – 2024

- Worked on a 3D CAD framework library in C#, focusing on geometric modeling and performance. Meshing of NURBS surfaces and B-Reps (video), mesh editing (video), CNC simulation and toolpath generation.
- Developed a debugging tool for 3D shapes called Blink.

ECMWF Summer of Weather Code *ECMWF, Reading* 2019

- Extended the Blender add-on BVTK for processing and visualization of meteorological data. Animations were displayed at the 2019 AGU (American Geophysical Union) meeting.

Python Developer for Scientific Computing *CINECA, Bologna* 2018 – 2019

- Developed a Blender add-on (BVTK) integrating scientific visualization via the VTK library.
- Presented at the Blender Conference in 2018.

Education

MSc Computer Science *ETH Zürich* 2024 – Present

- Major in Visual and Interactive Computing, minor in Machine Learning.

Overseas Exchange Program *University of Technology Sydney* 2022

BSc Computer Engineering *University of Bologna* 2019 – 2022

- Thesis: *Tessellation of boundary represented CAD models (B-Rep)*.
- Graduated with honors and won two scholarships for academic merit in 2019 and 2020.

Selected Projects

Semester Project *Computational Robotics Lab, ETH Zürich* 2026

- Developed Vorotile, a Voronoi-based generator of tileable patterns, supporting characterization of the mechanical properties of the resulting microstructures via homogenization analysis. Implemented in Python with JAX, NumPy, and Matplotlib. Web demo with vanilla WebGL.

Implicit SPH *Physically Based Simulations for Computer Graphics, Course Project* 2025

- Implemented in C++ an implicit Smoothed Particle Hydrodynamics (SPH) solver for elastic solids simulation, based on the paper "An Implicit SPH Formulation for Incompressible Linearly Elastic Solids" by Peer et al. (repository: github.com/lorenzocelli/implicit-sph).

Technical Skills

Programming Languages	Python, C, C++, C#, Java, Rust, Julia
Web Frameworks	React, React Native, Next.js, Tailwind, Django, Django REST Framework
Tools	Git, Docker, Linux, LaTeX

Language Skills

Italian Native **English** Fluent (C1)